The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Residual-Source Law
from a Centered Hidden-Response
Source-Balance Law

Abstract

The current active-primary theorem already proves that the current centered hidden-response residual law is the canonical quotient minimal-source-energy law of a smaller centered hidden-response residual-source law on the fixed displacement fibers. The next honest benchmark-facing move is therefore no longer another replay of that centered residual law, of the hidden-response Hessian minimization package, of the square-root operator core, of the normalized-lift package, of the body-level forcing law, or of the visible-response structures that follow from them. It is to derive or justify the centered residual-source law itself from still more primitive explicit substrate structure.

The present note takes that next step in the smallest clean balance-law form currently available below the centered-hidden-response-residual-source frontier. For each sector, it introduces a benchmark-faithful centered hidden-response source-balance law: the fixed proto-section benchmark base and fixed section-centered displacement body together with a finite-dimensional Euclidean balanced source carrier, a surjective balance operator onto a finite-dimensional balance target, a smooth surjective source displacement readout to the actual displacement fibers, and a smooth positive self-adjoint source-response operator. The current centered hidden-response residual-source law is then no longer primitive. Its residual state space is the balanced kernel

$$\mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}} = \ker B_\sigma^{\mathrm{disp,hid,res,src,bal}},$$

its source-state projection is the canonical response-orthogonal projector

$$P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) = I_{\mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}} - H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1} B_\sigma^{\mathrm{disp,hid,res,src,bal}*} \Big(B_\sigma^{\mathrm{disp,hid,res,src,bal}} H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y$$

and the induced source-state projection is exactly the unique balanced-kernel correction minimizing the fiberwise source-response energy

$$\mathcal{D}_\sigma^{\mathrm{disp,hid,res,src,bal}}(y; w, u) = \frac{1}{2} \left\langle w - u, H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)(w - u) \right\rangle .$$

Once that source-balance law is fixed, the current centered hidden-response residual-source law, the current centered hidden-response residual law, the current hidden-response Hessian minimization law, the same square-root operator core, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family, the same canonical minimal visible-response representative, the same current body-response core, and every currently recorded downstream benchmark-facing consequence follow unchanged.

The benchmark boundary remains unchanged throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The remaining honest burden therefore moves below the separate centered hidden-response residual-source law itself to a deeper derivation or justification of the centered hidden-response source-balance law that forces it.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has already moved one layer below the square-root-operator-core frontier, then one layer below the hidden-response-Hessian frontier, then one layer below the centered-residual frontier itself. The current active-primary theorem shows that the current centered hidden-response residual law is already the canonical quotient minimal-source-energy law of a smaller centered hidden-response residual-source law on the fixed displacement fibers.

The next honest question is therefore narrower again. What still more primitive explicit substrate structure forces that current centered hidden-response residual-source law itself? The present note answers that question in the smallest clean balance-law form now available below the current frontier. It does not reinstall the centered residual law, the hidden-response Hessian minimization law, the square-root operator core, the normalized-lift package, the selected-lift package, the body-law presentation, the visible-response operator family, the canonical minimal visible-response representative, the current body-response core, or any larger current-image-core or reduced-carrier or projection/body or full substrate-law presentation as primitive data. Instead it keeps only a balanced source carrier, a balance operator cutting out the balanced source kernel, a source displacement readout, and a positive source-response operator. The current centered residual-source law then descends canonically because its source-state projection is already forced as the response-orthogonal projector onto that balanced kernel.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim a completed first-principles derivation of GR from final substrate microphysics. Its narrower positive role is to show that the current centered hidden-response residual-source law already follows canonically from a smaller centered hidden-response source-balance law on the fixed displacement fibers.

2. Fixed Inputs from the Current Residual-Source Frontier

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded benchmark base and downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active route already fixes the following data and consequences.*

1. *the current proto-section benchmark base and in particular the recorded proto-section body* $\mathcal{B}_\sigma^{\mathrm{psec}}$ *and shell* $\mathcal{Z}_\sigma^{\mathrm{psec}}$*;*

2. *the fixed section-centered displacement body*

$$\mathcal{D}_\sigma^{\mathrm{pppresec}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{pppresec}}$$

whose fiber

$$\mathcal{D}_\sigma^{\mathrm{pppresec}}(y) \equiv \{v \in V_\sigma^{\mathrm{pppresec}} : (y, v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}\}$$

is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*;*

3. *the recorded downstream continuation*

$$\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec,res}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec}},$$
$$\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec,sec}},$$

together with the current continuity-Hessian compressed core and benchmark one-metric observer package;

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Definition 3 (Benchmark-faithful centered hidden-response residual-source law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* benchmark-faithful centered hidden-response residual-source law *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*

2. *A finite-dimensional Euclidean residual-source carrier*

$$\mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}.$$

3. *A finite-dimensional Euclidean residual state space*

$$\mathcal{U}_\sigma^{\mathrm{disp,hid,res}}.$$

4. *A smooth family of surjective linear source-state projections*

$$Q_\sigma^{\mathrm{disp,hid,res,src}}(y) : \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}} \to \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

5. *A smooth family of surjective linear source displacement readouts*

$$\Theta_\sigma^{\mathrm{disp,hid,res,src}}(y) : \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}} \to V_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

6. *Exactness of the displacement readout on source-state fibers:*

$$\ker Q_\sigma^{\mathrm{disp,hid,res,src}}(y) \subseteq \ker \Theta_\sigma^{\mathrm{disp,hid,res,src}}(y) \qquad \textit{for every } y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

7. *A smooth family of positive self-adjoint residual-source response operators*

$$K_\sigma^{\mathrm{disp,hid,res,src}}(y) : \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}} \to \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

   each of which is positive definite on $\mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}$.

8. *Benchmark faithfulness in the sense of Definition 2(4).*

Proposition 1 (What the current residual-source frontier already proves)**.** *At current scope, the already-recorded centered-hidden-response-residual-source frontier theorems prove all of the following.*

1. *Any benchmark-faithful centered hidden-response residual-source law canonically determines the current centered hidden-response residual law on the fixed displacement fibers.*

2. *That centered residual law canonically determines the current hidden-response Hessian minimization law.*

3. *That hidden-response Hessian minimization law canonically determines the current square-root operator core, the current section-centered displacement-body quadratic forcing law, the visible-response operator family* $S_\sigma^{\mathrm{disp,resp,vis}}$*, and the canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$.

4. *The current body-response core and every currently recorded downstream benchmark-facing consequence then follow unchanged.*

Proof. Item (1) is exactly the content of the current theorem proving that the centered hidden-response residual law is the quotient minimal-source-energy law of a smaller centered hidden-response

residual-source law. Items (2)–(4) are the already-recorded downstream continuation from that same residual-source frontier through the later centered-residual and hidden-response-Hessian and square-root-core consequences. □

3. Centered Hidden-Response Source-Balance Law

Definition 4 (Benchmark-faithful centered hidden-response source-balance law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* benchmark-faithful centered hidden-response source-balance law *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*
2. *A finite-dimensional Euclidean balanced source carrier*
$$\mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$
3. *A finite-dimensional Euclidean source-balance target*
$$\mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$
4. *A surjective linear source-balance operator*
$$B_\sigma^{\mathrm{disp,hid,res,src,bal}} : \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}} \to \mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$
5. *A smooth family of surjective linear source displacement readouts*
$$\Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) : \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}} \to V_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$
6. *A smooth family of positive self-adjoint source-response operators*
$$H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) : \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}} \to \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$
each of which is positive definite on $\mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}$.
7. *Response-orthogonal balance corrections are readout-invisible:*
$$H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1} B_\sigma^{\mathrm{disp,hid,res,src,bal}*} \mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}} \subseteq \ker \Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) \qquad \textit{for every } y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$
8. *Benchmark faithfulness in the sense of Definition 2(4).*

Remark 1. *Definition 4 is intentionally smaller at benchmark scope than the current centered hidden-response residual-source law. It does not install a separately primitive residual state space and does not install a separately primitive source-state projection. Instead the residual state space will be the balanced kernel of the explicit balance operator* $B_\sigma^{\mathrm{disp,hid,res,src,bal}}$*, and the source-state projection will be forced by* $B_\sigma^{\mathrm{disp,hid,res,src,bal}}$ *together with the positive source-response operator* $H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)$*. It also does not reinstall the centered residual law, the hidden-response Hessian minimization law, the square-root operator core, the body-level forcing law, the visible-response*

operator family, the canonical minimal visible-response representative, the current body-response core, or any larger same-output presentation as primitive data.

4. Induced Centered Hidden-Response Residual-Source Law

Definition 5 (Balanced kernel and response-orthogonal balance projector)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *and a benchmark-faithful centered hidden-response source-balance law. Define the balanced source state space*

$$\mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}} \equiv \ker B_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$

For $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*, define the source-balance quotient metric*

$$\Gamma_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) \equiv B_\sigma^{\mathrm{disp,hid,res,src,bal}} H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1} B_\sigma^{\mathrm{disp,hid,res,src,bal}^*}. \tag{2}$$

Also define the response-orthogonal balance corrector

$$\Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) \equiv H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1} B_\sigma^{\mathrm{disp,hid,res,src,bal}^*} \Gamma_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1} : \mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}} \to \mathcal{Y}_\sigma^{\mathrm{disp,hid,re} \tag{3}$$

and the balanced projection

$$P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) \equiv I_{\mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}} - \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) B_\sigma^{\mathrm{disp,hid,res,src,bal}}. \tag{4}$$

Finally, for $w \in \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}$ *and* $u \in \mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}}$*, define the source-balance correction energy*

$$\mathcal{D}_\sigma^{\mathrm{disp,hid,res,src,bal}}(y; w, u) \equiv \frac{1}{2} \left\langle w - u, H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)(w - u) \right\rangle.$$

Proposition 2 (The source-balance quotient metric is positive definite)**.** *For every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*, the operator* $\Gamma_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)$ *of* (2) *is positive self-adjoint and positive definite on* $\mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}}$*.*

Proof. Self-adjointness is immediate from the definition. For any $c \in \mathcal{C}_\sigma^{\mathrm{disp,hid,res,src,bal}}$,

$$\left\langle c, \Gamma_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) c \right\rangle = \left\langle H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1/2} B_\sigma^{\mathrm{disp,hid,res,src,bal}^*} c, H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1/2} B_\sigma^{\mathrm{disp,hid,res,src,bal}^*} c \right.$$

The right-hand side is nonnegative and vanishes only if

$$B_\sigma^{\mathrm{disp,hid,res,src,bal}^*} c = 0.$$

Because $B_\sigma^{\mathrm{disp,hid,res,src,bal}}$ is surjective, its adjoint is injective, so $B_\sigma^{\mathrm{disp,hid,res,src,bal}^*} c = 0$ implies $c = 0$. Hence $\Gamma_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)$ is positive definite. □

Proposition 3 (The balanced projection is the canonical projector onto the balanced kernel)**.** *For every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*, the map* $P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)$ *of* (4) *is a surjective idempotent linear map onto* $\mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}}$*. More precisely,*

1. $$B_\sigma^{\mathrm{disp,hid,res,src,bal}} P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) = 0;$$

2. $$P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) u = u \qquad \textit{for every } u \in \mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}};$$

3.

$$P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^2 = P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y);$$

4.

$$\ker P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) = \operatorname{im} \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y).$$

Proof. Because

$$B_\sigma^{\mathrm{disp,hid,res,src,bal}} \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) = B_\sigma^{\mathrm{disp,hid,res,src,bal}} H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1} B_\sigma^{\mathrm{disp,hid,res,src,bal}*} \Gamma_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)$$

we get

$$B_\sigma^{\mathrm{disp,hid,res,src,bal}} P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) = B_\sigma^{\mathrm{disp,hid,res,src,bal}} - B_\sigma^{\mathrm{disp,hid,res,src,bal}} \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) B_\sigma^{\mathrm{disp,hid,res,src,bal}} = 0.$$

Hence $\operatorname{im} P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) \subseteq \mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}}$.

If $u \in \mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}}$, then $B_\sigma^{\mathrm{disp,hid,res,src,bal}} u = 0$, so

$$P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) u = u - \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) B_\sigma^{\mathrm{disp,hid,res,src,bal}} u = u.$$

Therefore $P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)$ is surjective onto $\mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}}$, and idempotence follows:

$$P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^2 w = P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)\big(P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) w\big) = P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) w.$$

If $w \in \ker P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)$, then

$$w = \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) B_\sigma^{\mathrm{disp,hid,res,src,bal}} w \in \operatorname{im} \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y).$$

Conversely, if $w = \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) c$, then

$$P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) w = \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) c - \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) B_\sigma^{\mathrm{disp,hid,res,src,bal}} \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) c = 0.$$

Thus $\ker P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) = \operatorname{im} \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)$. □

Proposition 4 (Minimal source-balance correction characterization)**.** *For every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ and every $w \in \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}$, the element $P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) w$ is the unique vector in $\mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}}$ minimizing the source-balance correction energy:*

$$\mathcal{D}_\sigma^{\mathrm{disp,hid,res,src,bal}}(y; w, P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) w) = \min_{u \in \mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}}} \mathcal{D}_\sigma^{\mathrm{disp,hid,res,src,bal}}(y; w, u). \tag{5}$$

Proof. Fix $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ and $w \in \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}$. By Proposition 3, the vector $P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) w$ lies in $\mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}}$. Any other $u \in \mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}}$ can be written uniquely as

$$u = P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) w + \eta, \qquad \eta \in \mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}}.$$

Hence

$$w - u = \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) B_\sigma^{\mathrm{disp,hid,res,src,bal}} w - \eta.$$

Using

$$H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)\Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) = B_\sigma^{\mathrm{disp,hid,res,src,bal}*}\Gamma_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1},$$

we obtain

$$\begin{aligned}\left\langle \Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)B_\sigma^{\mathrm{disp,hid,res,src,bal}}w, H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)\eta\right\rangle &= \left\langle \Gamma_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)^{-1}B_\sigma^{\mathrm{disp,hid,res,src,bal}}w, B_\sigma^{\mathrm{disp,hi}}\right. \\ &= 0,\end{aligned}$$

because $\eta \in \ker B_\sigma^{\mathrm{disp,hid,res,src,bal}}$. Therefore

$$\begin{aligned}\mathcal{D}_\sigma^{\mathrm{disp,hid,res,src,bal}}(y; w, u) &= \mathcal{D}_\sigma^{\mathrm{disp,hid,res,src,bal}}(y; w, P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)w) \\ &\quad + \frac{1}{2}\left\langle \eta, H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)\eta\right\rangle.\end{aligned}$$

The second term is nonnegative and vanishes only when $\eta = 0$ because $H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)$ is positive definite. This proves (5) and uniqueness of the minimizer. □

Proposition 5 (The source displacement readout is exact on balanced-source fibers)**.** *Under Definition 4, for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$,*

$$\Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) = \Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y). \tag{6}$$

Equivalently,

$$\ker P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) \subseteq \ker \Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y).$$

Proof. By Definition 4(7),

$$\Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)\Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) = 0.$$

Hence, using (4),

$$\Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) = \Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y) - \Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)\Lambda_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)B_\sigma^{\mathrm{disp,hid,res,src}}$$

The kernel statement follows from Proposition 3(4). □

Definition 6 (Source-balance-induced centered residual-source law)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ and a benchmark-faithful centered hidden-response source-balance law. Define the induced centered hidden-response residual-source data by*

$$\mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}} \equiv \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}, \qquad \mathcal{U}_\sigma^{\mathrm{disp,hid,res}} \equiv \mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}},$$

$$Q_\sigma^{\mathrm{disp,hid,res,src}}(y) \equiv P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y), \qquad \Theta_\sigma^{\mathrm{disp,hid,res,src}}(y) \equiv \Xi_\sigma^{\mathrm{disp,hid,res,src,bal}}(y), \qquad K_\sigma^{\mathrm{disp,hid,res,src}}(y) \equiv H_\sigma^{\mathrm{disp}}$$

Proposition 6 (Source-balance-induced centered hidden-response residual-source law)**.** *Under Definitions 4 and 6, the data*

$$\left(\mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}, \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}, Q_\sigma^{\mathrm{disp,hid,res,src}}, \Theta_\sigma^{\mathrm{disp,hid,res,src}}, K_\sigma^{\mathrm{disp,hid,res,src}}\right)$$

form a benchmark-faithful centered hidden-response residual-source law in the sense of Definition 3.

Proof. The carrier $\mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}} = \mathcal{Y}_\sigma^{\mathrm{disp,hid,res,src,bal}}$ is finite-dimensional and Euclidean. The state space $\mathcal{U}_\sigma^{\mathrm{disp,hid,res}} = \mathcal{Z}_\sigma^{\mathrm{disp,hid,res,src,bal}}$ is a linear subspace of a finite-dimensional Euclidean space and is therefore finite-dimensional and Euclidean.

By Proposition 3, each $Q_\sigma^{\mathrm{disp,hid,res,src}}(y) = P_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)$ is a surjective linear projection onto $\mathcal{U}_\sigma^{\mathrm{disp,hid,res}}$. By Definition 4(5), each $\Theta_\sigma^{\mathrm{disp,hid,res,src}}(y)$ is a surjective linear map to the fixed displacement fibers. Proposition 5 gives the exactness condition

$$\ker Q_\sigma^{\mathrm{disp,hid,res,src}}(y) \subseteq \ker \Theta_\sigma^{\mathrm{disp,hid,res,src}}(y).$$

The operator $K_\sigma^{\mathrm{disp,hid,res,src}}(y) = H_\sigma^{\mathrm{disp,hid,res,src,bal}}(y)$ is positive self-adjoint and positive definite by Definition 4(6). Benchmark faithfulness is inherited directly from Definition 4(8). Therefore all items of Definition 3 hold. □

5. Deriving the Current Centered-Residual-Source Frontier

Theorem 1 (Current centered hidden-response residual-source law from a centered hidden-response source-balance law)**.** *Fix the primitive continuation data of Definition 1, the recorded benchmark base and downstream chain of Definition 2, and for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ a benchmark-faithful centered hidden-response source-balance law in the sense of Definition 4. Then the current centered hidden-response residual-source law is canonically forced by that smaller explicit substrate package. More precisely:*

1. *Proposition 6 canonically associates to the source-balance law a benchmark-faithful centered hidden-response residual-source law*

$$\left(\mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}, \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}, Q_\sigma^{\mathrm{disp,hid,res,src}}, \Theta_\sigma^{\mathrm{disp,hid,res,src}}, K_\sigma^{\mathrm{disp,hid,res,src}}\right).$$

2. *The induced source-state projection is exactly the response-orthogonal projector onto the balanced kernel:*

$$Q_\sigma^{\mathrm{disp,hid,res,src}}(y) = I_{\mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}} - K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1} B_\sigma^{\mathrm{disp,hid,res,src,bal}*} \Big(B_\sigma^{\mathrm{disp,hid,res,src,bal}} K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1}$$

and for every $w \in \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}$,

$$Q_\sigma^{\mathrm{disp,hid,res,src}}(y) w = \arg \min_{u \in \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}} \frac{1}{2} \left\langle w - u, K_\sigma^{\mathrm{disp,hid,res,src}}(y)(w-u) \right\rangle.$$

3. *By Proposition 1, the current centered hidden-response residual law, the current hidden-response Hessian minimization law, the same square-root operator core, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family $S_\sigma^{\mathrm{disp,resp,vis}}$, the same canonical minimal visible-response representative $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$, the same current body-response core, and every currently recorded downstream benchmark-facing consequence then follow unchanged from that same centered hidden-response source-balance law through the induced centered hidden-response residual-source law.*

4. *The benchmark one-metric package* (1) *is preserved, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

5. *Consequently the remaining honest burden is no longer to restate the current centered hidden-response residual-source law itself, but to derive or justify the centered hidden-response source-balance law that forces it from still more primitive explicit substrate structure.*

Proof. Item (1) is Proposition 6. Item (2) is exactly Propositions 3 and 4 rewritten in the induced residual-source notation of Definition 6. Item (3) is the already-recorded downstream continuation of the current residual-source frontier applied to the law furnished by Items (1)–(2). Item (4) is the benchmark-faithfulness condition built into Definitions 2 and 4. Item (5) is the routing consequence of Items (1)–(4): once the current centered hidden-response residual-source law is shown to be the canonical balanced-kernel output of the smaller centered hidden-response source-balance package, another manuscript that merely restates that same residual-source law or merely replays the same centered residual law or same hidden-response-Hessian or square-root-core or normalized-lift or selected-lift or body-law or visible-response consequences does not remove any remaining benchmark-facing freedom. □

Corollary 1 (The centered hidden-response residual-source law is no longer primitive)**.** *Under the hypotheses of Theorem 1, the current centered hidden-response residual-source law is no longer a primitive stopping point on the recorded section-centered hidden-response route. Its benchmark-relevant content is already forced by the smaller centered hidden-response source-balance law* $\mathfrak{B}^{\mathrm{disp,hid,res,src,bal}}$.

Proof. Theorem 1(1)–(3) identifies the current centered hidden-response residual-source law and its same centered-residual and hidden-response-Hessian and downstream benchmark-facing consequences as outputs of the smaller centered hidden-response source-balance law. Therefore the current residual-source package is no longer primitive at benchmark scope once that source-balance law is fixed. □

6. Conclusion

The positive result of this note is a genuine derivational gain below the current centered-hidden-response-residual-source frontier. The current centered hidden-response residual-source law no longer needs to be treated as primitive. It is already the canonical response-orthogonal balanced-kernel law induced by a smaller centered hidden-response source-balance package on the fixed displacement fibers.

The scope remains narrow and honest. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and nothing here upgrades adoption to a completed first-principles derivation of GR from final substrate microphysics.

The next open burden is therefore sharper again. The next benchmark-facing manuscript should derive or justify the centered hidden-response source-balance law itself from still more primitive explicit substrate structure. Another manuscript that merely restates the same centered hidden-response source-balance law, the same centered hidden-response residual-source law, the same centered hidden-response residual law, the same hidden-response Hessian minimization law, the same square-root operator core, the same carrier-bearing normalized selected-response lift, the same operator-bearing selected-lift package, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family, the same canonical minimal visible-response representative, the same current body-response core, or a larger same-output current-image-core or reduced-carrier or projection/body or full substrate-law presentation is no longer the clean next target at current scope.